Hutch++: Optimal Stochastic Trace Estimation

Meyer, Musco, Musco, and Woodruff. SOSA '21.

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Implicit Trace Estimation

Input: Access to an oracle that can evaluate $\boldsymbol{Ax}$ for any $\boldsymbol{x} \in \mathbb{R}^d$, where $\boldsymbol{A} \in \mathbb{R}^{d \times d}$

Output: Approximate $\mathrm{tr}(\boldsymbol{A})$ with as few queries to this oracle as possible

- Can get exact solution with d queries by $\mathrm{tr}(\boldsymbol{A}) = \sum_{i=1}^{d} \boldsymbol{e}_i^\top \boldsymbol{A} \boldsymbol{e}_i$
- Arises when $\boldsymbol{A}$ is a transformation of some other matrix $\boldsymbol{B}$
- e.g. $\boldsymbol{A} = \boldsymbol{B}^q, \boldsymbol{A} = \boldsymbol{B}^{-1}, \boldsymbol{A} = \exp(\boldsymbol{B})$
- Used to approximate matrix norms, spectral densities, log-determinants, Estrada index, eigenvalue counts in intervals, triangle counts in graphs, among others

Hutchinson's Trace Estimator

Algorithm $\mathrm{H}_m(\boldsymbol{A})$

$Z_0 \leftarrow 0$
for $i = 1$ to m **do**
  Generate $\boldsymbol{x} \in \mathbb{R}^d$ with i.i.d. entries with mean 0 and variance 1
  $Z \leftarrow \boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x}$ $\triangleright\ \mathbb{E}\, Z = \sum_{i,j} (\mathbb{E}\, \boldsymbol{x}\boldsymbol{x}^\top \odot \boldsymbol{A})_{i,j} = \mathrm{tr}(\boldsymbol{A})$
  $Z_0 \leftarrow Z_0 + \frac{1}{m} Z$
return Z_0

Hutchinson's Trace Estimator

Let entries of $\boldsymbol{G} = [\boldsymbol{g}_1, \ldots, \boldsymbol{g}_m] \in \boldsymbol{R}^{d \times m}$ have mean 0 and variance 1

$$\mathrm{H}_m(\boldsymbol{A}) = \frac{1}{m} \sum_{i=1}^{m} \boldsymbol{g}_i^\top \boldsymbol{A} \boldsymbol{g}_i^\top = \frac{1}{m} \operatorname{tr}(\boldsymbol{G}^\top \boldsymbol{A} \boldsymbol{G})$$

Theorem

When A is positive semidefinite, we can guarantee

$$(1-\varepsilon) \operatorname{tr}(\boldsymbol{A}) \leq \mathrm{H}_m(\boldsymbol{A}) \leq (1+\varepsilon) \operatorname{tr}(\boldsymbol{A})$$

with probability at least $1-\delta$ if we use $m = O(\log(1/\delta)/\varepsilon^2)$ mat-vec multiplication queries.

$O(1/\varepsilon^2)$ mat-vec mults for a $(1 \pm \varepsilon)$ guarantee

Looking under the hood

Theorem

With constant probability,

$$|\,\mathrm{H}_{O(1/\varepsilon^2)}(\boldsymbol{A}) - \mathrm{tr}(\boldsymbol{A})| \leq \varepsilon \|\boldsymbol{A}\|_F$$

- For a PSD matrix $\boldsymbol{A}$ with eigenvalues $\boldsymbol{\lambda}$, we have

$$\|\boldsymbol{A}\|_F = \|\boldsymbol{\lambda}\|_2 \leq \|\boldsymbol{\lambda}\|_1 = \mathrm{tr}(\boldsymbol{A})$$

- Bound is only tight when $\boldsymbol{A}$ has significant mass concentrated on just a small number of eigenvalues.

Key Idea: Project off top eigenvalues!

Algorithm Sketch

- Let $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ be PSD, and let m be budget of $\boldsymbol{A}\boldsymbol{x}$ queries
- Compute orthonormal span $\boldsymbol{Q} \in \mathbb{R}^{d \times \frac{m}{3}}$ of $\boldsymbol{A}$'s top eigenvectors
- Separate $\boldsymbol{A}$ into projection onto its susbpace spanned by $\boldsymbol{Q}$ and onto that subspace's orthogonal compliment

$$\operatorname{tr}(\boldsymbol{A}) = \underbrace{\operatorname{tr}\left(\boldsymbol{Q}\boldsymbol{Q}^\top \boldsymbol{A}\boldsymbol{Q}\boldsymbol{Q}^\top\right)}_{= \underbrace{\operatorname{tr}\left(\boldsymbol{Q}^\top \boldsymbol{A}\boldsymbol{Q}\boldsymbol{Q}^\top \boldsymbol{Q}\right)}_{= \operatorname{tr}(\boldsymbol{Q}^\top \boldsymbol{A}\boldsymbol{Q})}} + \operatorname{tr}\left(\underbrace{(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)}_{\|\boldsymbol{\Delta}\|_F^2 \leq \varepsilon \operatorname{tr}(\boldsymbol{A})^2}\right)$$

- Compute $\operatorname{tr}(\boldsymbol{Q}^\top \boldsymbol{A}\boldsymbol{Q})$ exactly
- Use $\mathrm{H}_{\frac{m}{3}}(\boldsymbol{\Delta})$ to approx remaining trace

Living the good life

- Suppose $\boldsymbol{Q} = \boldsymbol{V}_k$ is the orthonormal basis spanning top k eigenvectors of $\boldsymbol{A}$
- Then $\boldsymbol{A}_k = \boldsymbol{V}_k^\top \boldsymbol{A} \boldsymbol{V}_k$, $\boldsymbol{\Delta} = \boldsymbol{A} - \boldsymbol{A}_k$

Theorem

For any PSD matrix $\boldsymbol{A}$, $\|\boldsymbol{A} - \boldsymbol{A}_k\|_F \leq \frac{1}{\sqrt{k}} \operatorname{tr}(\boldsymbol{A})$

Proof.

- Let $\boldsymbol{A}$ have eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$
- $\lambda_{k+1} \leq \frac{1}{k} \sum_{i=1}^k \lambda_i \leq \frac{1}{k} \operatorname{tr}(\boldsymbol{A})$

$$\|\boldsymbol{A} - \boldsymbol{A}_k\|_F^2 = \sum_{i=k+1}^{d} \lambda_i^2 \leq \lambda_{k+1} \sum_{i=k+1}^{d} \lambda_i \leq \frac{1}{k} \operatorname{tr}(\boldsymbol{A})^2 \qquad \square$$

Living the good life

- Suppose $\boldsymbol{Q} = \boldsymbol{V}_k$ is the orthonormal basis spanning top k eigenvectors of $\boldsymbol{A}$
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Theorem

For any PSD matrix $\boldsymbol{A}$, $\|\boldsymbol{A} - \boldsymbol{A}_k\|_F \leq \frac{1}{\sqrt{k}} \operatorname{tr}(\boldsymbol{A})$

- Set $k = O(1/\varepsilon)$ and let $\operatorname{tr}(\boldsymbol{A}) = \operatorname{tr}(\boldsymbol{A}_k) + \operatorname{tr}(\boldsymbol{A} - \boldsymbol{A}_k)$
- Calculate $\operatorname{tr}(\boldsymbol{A}_k) = \operatorname{tr}(\boldsymbol{V}_k^\top \boldsymbol{A} \boldsymbol{V}_k)$ exactly
- Approximate $\operatorname{tr}(\boldsymbol{\Delta}) = \operatorname{tr}(\boldsymbol{A} - \boldsymbol{A}_k)$ with Hutchinson's estimator

$$\begin{aligned}
|\,\mathrm{H}_{O(1/\varepsilon)}(\boldsymbol{\Delta}) - \operatorname{tr}(\boldsymbol{\Delta})| &\leq \sqrt{\varepsilon}\|\boldsymbol{\Delta}\|_F \\
&\leq \sqrt{\frac{\varepsilon}{k}} \operatorname{tr}(\boldsymbol{A}) \\
&= O(\varepsilon \operatorname{tr}(\boldsymbol{A}))
\end{aligned}$$

Living la vida loca

- What to do without knowing $\boldsymbol{V}_k$?
- Goal is to find a matrix $\boldsymbol{Q}$ in $O(k)$ mat-vec queries s.t.

$$\|\boldsymbol{\Delta}\|_F = \|(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\|_F \leq O(\|\boldsymbol{A} - \boldsymbol{A}_k\|_F)$$

- Let $\boldsymbol{S} \in \mathbb{R}^{d \times k}$ be a random sign matrix
- Let $\boldsymbol{Q}$ be an orthonormal basis for the column span of $\boldsymbol{AS}$
- Then w.h.p. $\boldsymbol{Q}$ aligns with large eigenvectors of $\boldsymbol{A}$

$$\|(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\|_F^2 \leq \|\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{Q}\boldsymbol{Q}^\top)\|_F^2 \leq 2\|\boldsymbol{A} - \boldsymbol{A}_k\|_F^2$$

Hutch++

Algorithm 1 Hutch++

input: Matrix-vector multiplication oracle for matrix, $\boldsymbol{A} \in \mathbb{R}^{d\times d}$. Number of queries, m.
output: Approximation to $\mathrm{tr}(\boldsymbol{A})$.

1: Sample $\boldsymbol{S} \in \mathbb{R}^{d\times \frac{m}{3}}$ and $\boldsymbol{G} \in \mathbb{R}^{d\times \frac{m}{3}}$ with i.i.d. $\{+1, -1\}$ entries.
2: Compute an orthonormal basis $\boldsymbol{Q} \in \mathbb{R}^{d\times \frac{m}{3}}$ for the span of $\boldsymbol{AS}$ (e.g., via QR decomposition).
3: **return** Hutch++$(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{Q}^T\boldsymbol{AQ}) + \frac{3}{m}\mathrm{tr}(\boldsymbol{G}^T(\boldsymbol{I} - \boldsymbol{QQ}^T)\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{QQ}^T)\boldsymbol{G})$.

$O(1/\varepsilon)$ mat-vec mults for a $(1 \pm \varepsilon)$ guarantee

Extensions

The authors of [MMMW] also provide:

- a non-adaptive variant of Hutch++ using $O(1/\varepsilon)$ queries
- an $\Omega(\frac{1}{\varepsilon \log(1/\epsilon)})$ lower bound for queries of bounded bit complexity
- an $\Omega(1/\varepsilon)$ lower bound for non-adaptive algorithms in RAM model
- an upper bound on Hutch++ for non-PSD matrices
- variance analysis with small explicit constants
- empirical results!

Empirical Results

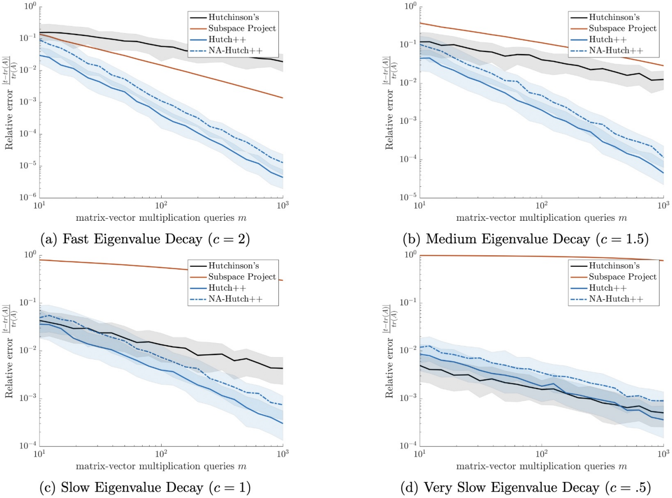
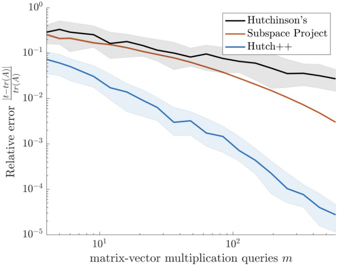
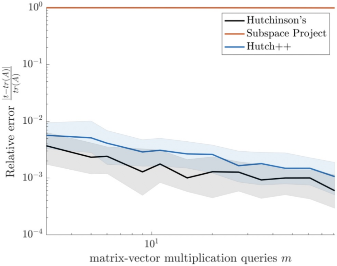


Figure 1: Relative error versus number of matrix-vector multiplication queries for trace approximation algorithms run on random matrices with power law spectra. We report the median relative

Empirical Results

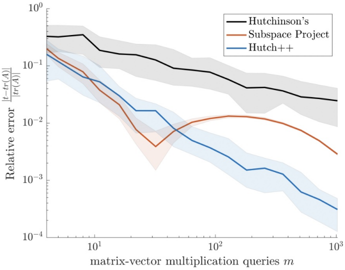


(a) $\boldsymbol{A} = \exp(\boldsymbol{B})$, where $\boldsymbol{B}$ is the Roget's Thesaurus semantic graph adjacency matrix. For use in Estrada index computation. $\boldsymbol{A}$ is PSD.

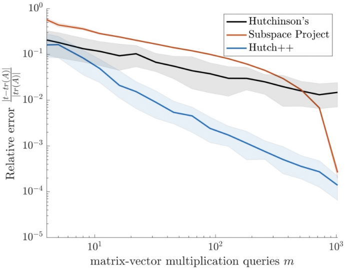


(b) $\boldsymbol{A} = \log(\boldsymbol{B} + \lambda \boldsymbol{I})$, where $\boldsymbol{B}$ is a 2D Gaussian process kernel covariance matrix. For use in log-likelihood computation. $\boldsymbol{A}$ is not PSD.

Empirical Results



(a) $\boldsymbol{A} = \boldsymbol{B}^3$ where $\boldsymbol{B}$ is a Wikipedia voting network adjacency matrix. For use in triangle counting. $\boldsymbol{A}$ is not PSD.



(b) $\boldsymbol{A} = \boldsymbol{B}^3$ where $\boldsymbol{B}$ is an arXiv.org citation network adjacency matrix. For use in triangle counting. $\boldsymbol{A}$ is not PSD.

References I

Raphael A. Meyer, Cameron Musco, Christopher Musco, and David P. Woodruff.
Hutch++: Optimal stochastic trace estimation.
In *2021 Symposium on Simplicity in Algorithms (SOSA)*, pages 142–155.